

BIS 580 Project Report

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Section 1: Summary of Techniques

Goal of Project: To estimate property prices based on structural factors (room number, age) and area characteristics (crime rate, air quality, proximity to airport, water bodies, rainfall, and bus stop).

Descriptive Methods Employed:

1. Summary Statistics: I calculated mean, median, standard deviation, range, etc. for all numerical variables
2. Pivot Tables: I analyzed average prices by airport proximity, waterbody type, bus stop, crime rate, rainfall, house age, and air quality
3. Pivot Chart: I analyzed average prices by airport proximity, room number, crime rate, rainfall, house age, and air quality
4. Data Visualization: I employed scatter plots of price vs. room number, and crime rate

Predictive Models Employed:

1. Multiple Linear Regression: I built a model with price as the dependent variable and all other features as predictors

Section 2: Discussion of the Techniques Used

Descriptive Analytics Rationale:

1. Summary statistics helped understand the distribution of house prices, revealing a right-skewed distribution with most houses priced between \$130,000-\$500,000
2. Scatter plots revealed clear positive relationships between room number and price, and negative relationships between crime rate and price.
3. Correlation analysis identified multicollinearity concerns (e.g., age and crime rate show some correlation, air quality and crime rate also showed some correlation, etc.)

Predictive Modeling Approach

Linear regression was chosen because house prices are a continuous variable, making regression an appropriate choice; the relationships appear reasonably linear based on scatter plots. This approach was used because I wanted to quantify the impact of each feature on the average house price.

All available features were analyzed, and the decision was based on the relationship between each variable of interest.

Section 3: Insights and Findings

Key Descriptive Findings

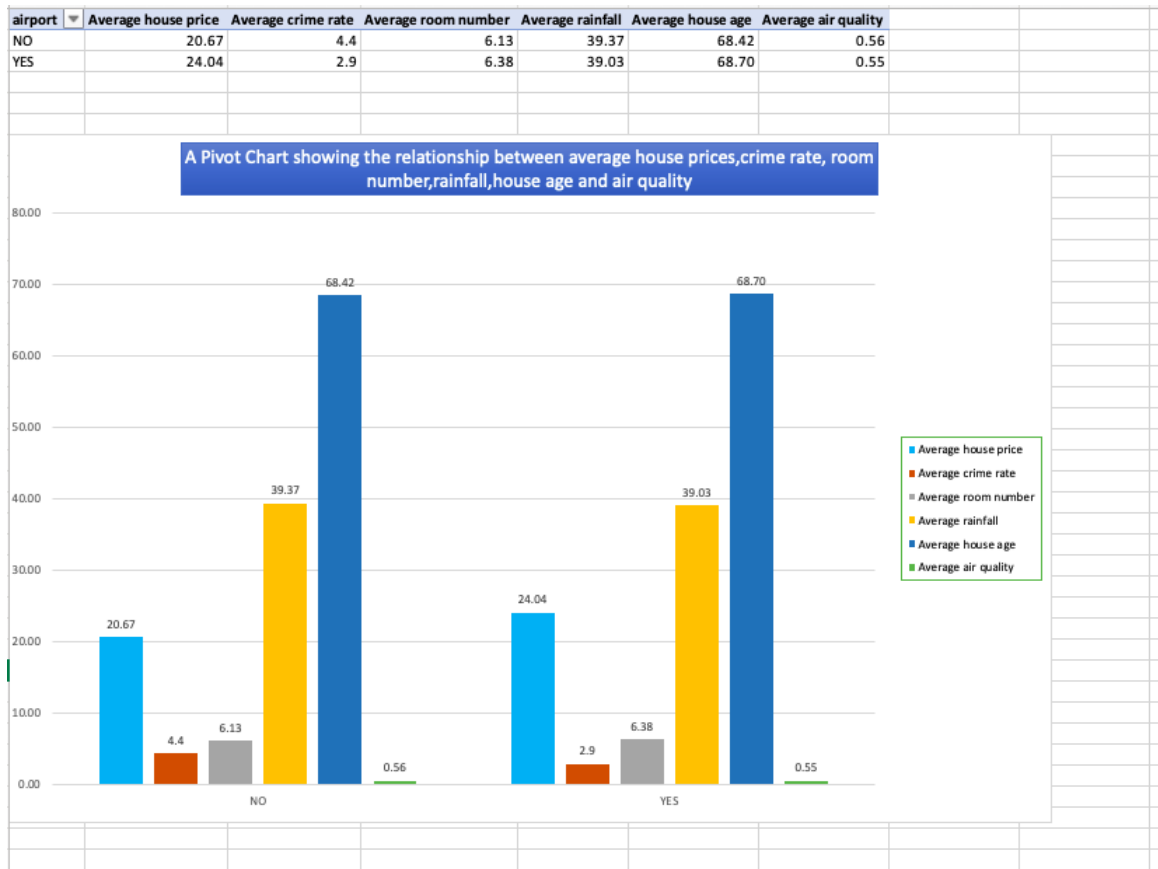


Figure 1

The following findings can be derived from the pivot chart above

1. Price Distribution: Average houses are priced between \$206,700-\$240,400, with a few luxury properties up to \$500,000
2. Room Impact: Houses with 6 or more rooms also average between \$206,700-\$240,400
3. Rainfall: Properties in areas with high rainfall command higher average house prices than those with no rainfall.
4. Crime Impact: Areas with crime rates above 4.0 (high) have lower average house prices, and areas with crime rates of 2.9 and below (low) have higher average house prices.
5. Airport Proximity: Airport proximity to properties showed a positive average house price impact and vice versa

Predictive Model Results:

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.744592296								
R Square	0.554417687								
Adjusted R Square	0.548154481								
Standard Error	6.172208623								
Observations	506								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	7	23605.8514	3372.26449	88.5198023	2.6465E-83				
Residual	498	18971.8873	38.0961593						
Total	505	42577.7387							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	-17.26348158	3.36570265	-5.1292355	4.1743E-07	-23.876209	-10.650754	-23.876209	-10.650754	
crime_rate	-0.205698112	0.03550241	-5.7939194	1.2204E-08	-0.2754511	-0.1359451	-0.2754511	-0.1359451	
air_quality	-6.99114157	3.66246226	-1.9088638	0.05685418	-14.186924	0.2046408	-14.186924	0.2046408	
room_num	7.233971507	0.40267877	17.964621	5.358E-56	6.44281283	8.02513019	6.44281283	8.02513019	
age	-0.033942382	0.0144424	-2.3501892	0.01915317	-0.0623179	-0.0055668	-0.0623179	-0.0055668	
rainfall	0.013122021	0.0220817	0.59424872	0.55261563	-0.0302628	0.05650679	-0.0302628	0.05650679	
Airport.a	1.193303527	0.56299833	2.11955076	0.03453806	0.08715877	2.29944828	0.08715877	2.29944828	
waterbody.a	0.339097679	0.59904631	0.56606255	0.57160624	-0.837872	1.51606731	-0.837872	1.51606731	
Regression equation									
Y (price)= -17.11+ (-0.21)*crime rate+(-6.99)*air quality+7.23(room number)+(-0.03)*age+(0.01)*rainfall+1.19*(rainfall)+0.33*(waterbody)									
The R ² value of 0.55 indicates that approximately 55% of the variation in house prices can be explained by the model's independent or explanatory variables (crime rate, air quality, room number, age,rainfall, airport and waterbody).									
Decision rule : If p-value <= 0.05 level of significance, we reject null hypothesis. This means the variable is statistically significant									
f p-value > 0.05 level of significance, we do not reject null hypothesis. This means the variable is not statistically significant									

Figure 2

Model Performance: R² = 0.55

From Figure 2, the R² value of 0.55 indicates that about 55% of the variation in house prices can be explained by the model's independent variables, including crime rate, air quality, number of rooms, age, rainfall, proximity to the airport, and waterbody. Dummy variables were used for both airport and waterbody.

The coefficient on the "crime rate" variable is -0.21, which means that, holding all other variables constant, a unit decrease in crime leads to an increase in house prices by \$2,100 (0.21*\$10,000). Hence an inverse relationship between average house prices and crime rate.

Since the p-value (0.00) ≤ 0.05 significance level, we reject the null hypothesis that crime rate has no significant effect on house prices. Hence, the crime rate has a significant effect on average house prices, holding all other factors constant.

The coefficient on the "air quality" variable is -7.48, meaning that for an additional increase in air quality, house prices are predicted to rise by \$74,800 (7.48*\$10,000), holding all variables constant.

Since the p-value (0.04) ≤ 0.05 significance level, we reject the null hypothesis that air quality does not affect house prices. Hence, average house prices could be influenced significantly by air quality.

The coefficient on the "room number" variable is 7.37, meaning that a 1 unit increase in the number of rooms leads to a rise in house prices by \$73,700 (7.37*\$10,000), holding all variables constant.

Since the p-value (0.00) ≤ 0.05 significance level, we reject the null hypothesis that room number does not affect house prices. Hence, average house prices are greatly influenced by an increase in the number of rooms in a house.

The coefficient on the "age" variable is -0.03, meaning that a one-year increase in the age of a house leads to a fall of \$300 ($0.03 \times \$10,000$) in house prices, holding all variables constant. This suggests that older houses are associated with lower prices and vice versa.

Since the p-value ($0.02 \leq 0.05$ significance level), we reject the null hypothesis that age has no significant effect on house prices. Hence, average house prices are greatly influenced by the increase in the age of a house.

The coefficient on the "rainfall" variable is 0.01, meaning that for every unit increase in rainfall, house prices are predicted to increase by \$100 ($0.01 \times \$10,000$), holding all variables constant.

Since the p-value ($0.55 > 0.05$ significance level), we fail to reject (accept) the null hypothesis that rainfall has no significant effect on house prices. Hence, house prices are influenced by other variables, holding all other factors constant.

The coefficient on the "airport" variable is 1.19, meaning that for every house that is closer to the airport, house prices are predicted to increase, holding all variables constant.

Since the p-value ($0.03 < 0.05$ significance level), we reject the null hypothesis that the airport has no significant effect on house prices. Hence, house prices are greatly influenced by proximity to the airport, holding all other factors constant.

The coefficient on the "waterbody" variable is 0.33, meaning that for every unit increase in waterbody presence, house prices are predicted to increase by \$3,300 ($0.33 \times \$10,000$), holding all variables constant.

Since the p-value ($0.57 > 0.05$ significance level), we fail to reject (accept) the null hypothesis that waterbody has no significant effect on house prices. Hence, house prices are influenced by other variables, holding all other factors constant.

Recommendation from findings:

First and foremost, developers should focus on adding rooms (highest ROI) and securing an environment with a low rate of rainfall and no water bodies to maximize profits. This is because rainfall and waterbodies have no significant effect on house prices.

Buyers should prioritize low-crime areas and better air quality for investment value. The government should also implement sound and proactive policies to mitigate crime in high-crime areas. This will boost investor confidence and drive sustainable growth within the crime zone areas or communities.

Continued focus on crime reduction strategies is crucial, as small decreases in crime rates can lead to significant increases in housing prices, indicating increased desirability and stability in communities.

Moreover, reducing crime and improving air quality can significantly increase property values and help in municipal planning. It is therefore prudent to invest in environmental initiatives to improve air quality.

The results from the model can also be used to set competitive prices based on feature combinations. This means businesses or developers can identify which features most influence customer demand and adjust pricing accordingly.

In a nutshell, the findings provide actionable insights for real estate stakeholders to make well-informed decisions to maximize profit and accelerate housing development within the economy.

References

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